

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

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About Me

I've spent 10 years building scalable backend systems and infrastructure, with the last 3 focused on technical leadership and architecture. I design systems that serve multiple teams: APIs processing \$4.3M in transactions with 99.9% uptime, microservices on Kubernetes handling millions of events, and developer platforms that reduced provisioning by 60x. I work in Python, Java, TypeScript, and C# on AWS with Postgres, Redis, and Elasticsearch. I've both led teams directly and led through technical vision via RFCs, mentorship, and code quality. I use AI tools (Cursor, ChatGPT, Copilot) daily to accelerate development while maintaining quality. I'm part architect, part builder, part mentor.

Work Experience

Alberta Motor Association

Edmonton, Alberta

SOFTWARE ARCHITECT

Jan 2023 - Present

- Built self-service developer platform solo that reduced environment provisioning from 2-3 weeks to 2 hours (60x improvement). Adopted by 15 engineers across 3 teams, created 120+ environments. Unblocked major initiatives by enabling rapid iteration and production debugging. (Python, Terraform, AWS, Jenkins, Docker)
- Own API architecture for platform serving 5 internal teams and external partners. Designed API contracts, approve endpoint additions, maintain OpenAPI standards, and ensure consistency across services. APIs process \$4.3M in annual transactions. (Python, REST, OpenAPI, TypeScript)
- Architected technical infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts. 10 Lambda functions, \$4.3M annual revenue, 2.8M events processed, 99.9% uptime. Won 2024 Guidewire Innovation Award. (Python, AWS Lambda, Terraform, TypeScript)
- Led modernization of document generation system from legacy C# to containerized microservices on EKS. Led 2-3 engineers. Integrated Datadog for observability (failure rates, success metrics, performance). Reduced environment provisioning from weeks to minutes. (Kubernetes, EKS, Docker, Datadog, Helm, C#)
- Set technical standards and provide architectural guidance for 12-person team through code reviews, RFC authorship, and technical documentation. Member of Architecture Steering Committee. Mentor engineers on system design, debugging, and best practices.

Arterys Inc. (Acquired by Tempus Labs)

Calgary, Alberta (Remote)

SOFTWARE DEVELOPER

Aug 2021 - Aug 2022

- Built backend services running medical AI algorithms, optimizing for low latency and high throughput. (TypeScript, Node.js, AWS)
- Improved algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, identifying and resolving issues before customer impact.

Alberta Motor Association

Edmonton, Alberta

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework from scratch that reduced defects by 30% on subsequent releases. Framework adopted org-wide, still in use 4+ years later across all development teams. Significantly improved developer velocity and release confidence. (Java/Gosu, JUnit)
- Led 5-6 person development team using TDD methodology. Delivered projects on time with no critical defects post-launch. Conducted code reviews, mentored junior developers, and collaborated with stakeholders on requirements and priorities.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local dev environments, reducing onboarding friction for new developers.

Pleasant Solutions

Edmonton, Alberta

SOFTWARE DEVELOPER

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD. (C#)

Lockheed Martin Canada

Halifax, Nova Scotia

SOFTWARE ENGINEER ASSOCIATE

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining real-time data streams from multiple sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system for capturing and replaying test scenarios, improving debugging and test coverage.

The King's University

Edmonton, Alberta

SOFTWARE DEVELOPER/RESEARCH ASSISTANT

May 2014 - Sep 2016

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.

Education

The Open University

MSC. COMPUTING (SOFTWARE ENGINEERING)

Currently pursuing Master's in Computing at The Open University.

Milton Keynes, England (Remote)

May 2024 - Ongoing

The King's University

B.S. IN COMPUTER SCIENCE

Edmonton, Alberta

Sep 2012 - Dec 2016

Skills

Technical Leadership & Force Multiplication

Lead cross-team initiatives and mentor engineers at all levels. Set standards for 12-person team through RFC authorship, code reviews, and architectural guidance. Member of Architecture Steering Committee. Champion engineering practices and reusable solutions. Balance technical vision with pragmatic delivery.

Scalable Systems & Architecture

Design highly scalable systems serving 5 teams with 99.9% uptime, handling millions of events and \$4.3M in transactions. Built multi-account AWS architectures, microservices on Kubernetes, and event-driven systems. Focus on maintainability, performance, and operational excellence.

Platform & Developer Experience

Built developer platforms (60x faster provisioning) and test frameworks (30% defect reduction) adopted org-wide. Design tooling that multiplies team effectiveness. Modernized legacy systems to cloud-native architectures. Drive innovation through best practices and emerging technologies.

Cross-Functional Collaboration

Partner with Product and Design to shape roadmap and deliver customer-facing features. Communicate complex technical concepts to diverse audiences. Navigate ambiguity and prioritize effectively in fast-paced environments. Bridge technical and non-technical stakeholders.

Technologies

Languages

Python, Java, TypeScript, C#, Bash

Cloud & Infrastructure

AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM), Docker, Kubernetes, Terraform, Helm

Databases

PostgreSQL, Aurora/RDS, Redis, Elasticsearch, SQL

Architecture

Microservices, Event-Driven Architecture, REST APIs, OpenAPI, Distributed Systems

Tools & Practices

Datadog, CI/CD, Git, Infrastructure as Code, AI-Powered Development (Cursor, ChatGPT, Copilot)